

R2 Scores of different ML models:

PROBLEM STATEMENT:

Profit prediction from 50 startups dataset

1.MULTIPLE LINEAR REGRESSION

R-Squared Score – 93%

2. SUPPORT VECTOR MACHINE

S.NO	KERNAL	C	R2 SCORE
1	Linear	1	0.9
2	Linear	0.1	0.92
3	Linear	0.01	0.93
4	Poly	1	-0.5
5	Poly	0.1	-0.6
6	Poly	0.01	-0.6
7	rbf	1	-0.3
8	rbf	0.1	-0.5
9	rbf	0.01	-0.6
10	sigmoid	1	-0.6
11	sigmoid	0.1	-0.6
12	sigmoid	0.01	-0.6

3. DECISION TREE

S.NO	SPLITTER	CRITERION	R2 SCORE
1	Absolute error	Best	0.95
2	Absolute error	random	0.89
3	Squared error	Best	0.91
4	Squared error	random	0.90
5	Friedman mse	Best	0.91
6	Friedman mse	random	0.91
7	Poisson	Best	0.90
8	Poisson	random	0.93

4. RANDOM FOREST

S.NO	N_ESTIMATORS	CRITERION	MAX_FEATURES	R2 SCORE
1	100	Squared error	sqrt	0.75
2	10	Squared error	sqrt	0.83
3	50	Squared error	sqrt	0.8
4	100	Squared error	log2	0.77
5	10	Squared error	log2	0.78
6	50	Squared error	log2	0.83
7	100	Absolute error	sqrt	0.81
8	10	Absolute error	sqrt	0.69
9	50	Absolute error	sqrt	0.78
10	100	Absolute error	log2	0.81
11	10	Absolute error	log2	0.65
12	50	Absolute error	log2	0.76
13	100	Friedman mse	sqrt	0.76
14	10	Friedman mse	sqrt	0.56
15	50	Friedman mse	sqrt	0.8
16	100	Friedman mse	log2	0.77
17	10	Friedman mse	log2	0.83
18	50	Friedman mse	log2	0.8
19	100	Poisson	sqrt	0.8
20	10	Poisson	sqrt	0.42
21	50	Poisson	sqrt	0.77
22	100	Poisson	log2	0.79
23	10	Poisson	log2	0.66
24	50	Poisson	log2	0.76